

Executive Summary

AAA Washington's emergency road service operation accounts for over one third of the organization's operating budget. Staffing too many dispatchers costs money. Staffing too few means members wait longer for help. Getting the forecast right matters.

This project set out to answer a straightforward question: what actually drives service call volume, and can we predict it reliably enough to plan around it?

What We Found

Call volume is seasonal and predictable. Demand climbs every winter and drops every summer without fail. But that alone does not explain why some years are significantly worse than others.

Two factors fill that gap.

Temperature is the strongest driver. Vehicle failures pick up noticeably when temperatures drop below 60 degrees. The further a month falls below that threshold, the more calls come in.

Unemployment is an early warning signal. When unemployment rises, people put off car maintenance. About 11 months later, those neglected vehicles start failing. That lag gives AAA Washington a head start on demand spikes using nothing more than publicly available economic data.

How Well Can We Forecast?

Two models were built and tested against historical data:

Model	Accuracy	Best Used For
Seasonal forecasting model	Within 2.5% of actual call volume	Annual budget and staffing planning
Temperature and unemployment model	Within 3.6% of actual call volume	In-season adjustments and early warning

Recommendations

Use the seasonal model for annual planning. Its accuracy is strong enough to support headcount decisions and budget commitments at the start of each year.

Use the temperature and unemployment model throughout the year. When unemployment starts climbing, or a cold winter is in the forecast, this model puts a number on how much demand is likely to increase and how soon. That gives operations time to prepare rather than scramble.

Both models should be updated quarterly as new data comes in. Adding weather severity or traffic incident data down the road could improve accuracy further.